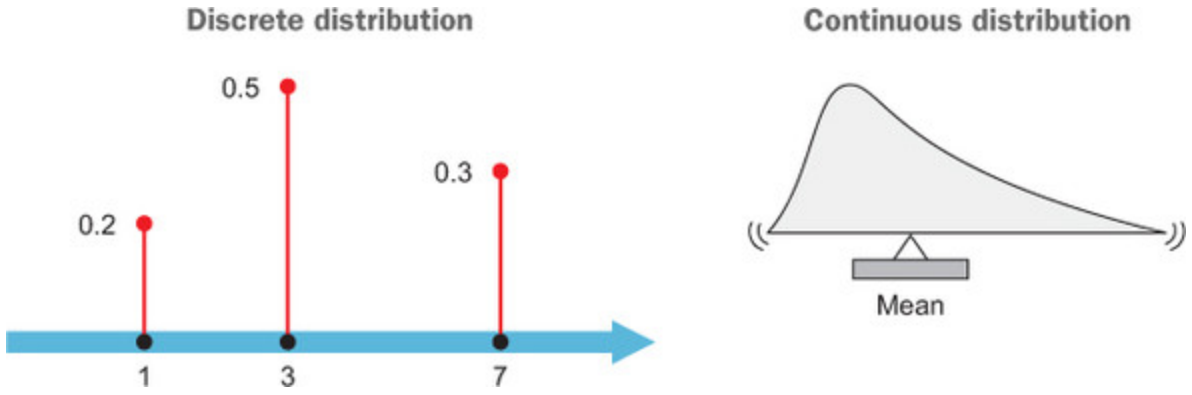
However, tomorrow's temperature could be any real number (or if measured in Kelvin, it could be any real number from 0 to infinity), and the real numbers or any subset of the real numbers is infinite since we can continually divide them: 1.5 is a real number, and so is 1.500001, and so forth. When the sample space is infinite, this is a *continuous probability distribution*.

In a continuous probability distribution, the distribution does not tell you the probability of a particular outcome, because with an infinite number of possible outcomes, each individual outcome must have an infinitely small probability in order for the sum to be 1. Thus, a continuous probability distribution tells you the *probability density* around a particular possible outcome. The probability density is the sum of probabilities around a small interval of some value—it's the probability that the outcome will fall within some small interval. The difference between discrete and continuous distributions is depicted in figure 7.6. That's all we'll say about continuous distributions for now because in this book we'll really only deal with discrete probability distributions.

Figure 7.6. Left: A discrete distribution is like a numpy array of probabilities associated with another numpy array of outcome values. There is a finite set of probabilities and outcomes. Right: A continuous distribution represents an infinite number of possible outcomes, and the y axis is the probability density (which is the probability that the outcome takes on a value within a small interval).



Another question we can ask of a probability distribution is its spread or *variance*. Our beliefs about something can be more or less confident, so a probability distribution can be narrow or wide respectively. The calculation of variance uses the expectation operator and is defined as